

Work

$$\vec{E}_{x_1} = \text{Work}$$

$$w = \vec{F} \cdot \vec{d} = (F_x \times d_x) + (F_y \times d_y) \\ = (3 \times 5) + (4 \times 2) \\ = 15 + 8 = 23 \text{ J.}$$

$$\vec{E}_{x_2}$$

$$\sum \vec{F} = \vec{0} \text{ (constant speed)}$$

Block (A)

$$\vec{N} + \vec{W}_A + \vec{F}_f + \vec{T} = \vec{0}$$

$$\Rightarrow N - W_A - F_f + T = 0. \Rightarrow \begin{cases} \text{We have } N = W_A = 20 \text{ N} \\ \therefore T = F_f = 12 \text{ N.} \end{cases}$$

Block (B)

$$W_B - T = 0 \Rightarrow W_B = T = 12 \text{ N}$$

a) Work done on block (B) (12 N).

$$\text{i) work done by gravity: } W = F \cdot d = m_B g \cdot 0,75 \cos 0 = 9 \text{ J}$$

$$\text{ii) " " " the tension: } W = T \cdot d \cos \pi = -12 \times 0,75 = -9 \text{ J}$$

b) Work done on block (A) (20 N)

$$\text{i) by gravity: } W = m_A g \times 0,75 \times \cos 90 = 0 \text{ J.}$$

$$\text{ii) by tension: } W = T \times 0,75 \cos 0 = 9 \text{ J.}$$

$$\text{iii) by friction: } W = F_f \times d \cos \pi = -12 \times 0,75 = -9 \text{ J}$$

$$\text{iv) by Normal force } W = N \times d \cos \frac{\pi}{2} = 0 \text{ J.}$$

c). total work done on each Block:

$$\text{block (A): } W_{\text{tot}} = 9 - 9 = 0 \text{ J.}$$

$$\text{block (B): } W_{\text{tot}} = 0 + 9 - 9 + 0 = 0 \text{ J}$$

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Ex 4

Work

- 1) the work done on the book between A and B is the change of its kinetic energy between A and B.

$$W = E_{K(B)} - E_{K(A)} = \frac{1}{2} m (v_B^2 - v_A^2) = \frac{1}{2} (1.5) ((1.25)^2 - (3.21)^2) = -6.55 \text{ J.}$$

- 2) Velocity at C if $w = -0.75 \text{ J}$.

$$W = \frac{1}{2} m v_C^2 - \frac{1}{2} m v_B^2 \Rightarrow v_C = \sqrt{\frac{2W}{m} + v_B^2}$$

$$\Rightarrow v_C = 0.75 \text{ m/s.}$$

- 3) Velocity at C. if $w = +0.75 \text{ J}$.

$$v_C = \sqrt{\frac{2W}{m} + v_B^2} = 1.6 \text{ m/s.}$$

Ex 5

- 1) Value of d

$$W = \vec{F} \cdot \vec{d} = F \cdot d \cdot \cos \theta$$

$$F = \vec{W} + \vec{N} + \vec{F}_f = \vec{F}_f \quad (\vec{W} = -\vec{N}) \quad (W = N) \quad \text{and } \theta = \pi. \quad / F_f = \mu_k N = \mu_k mg$$

$$W = F_f \cdot d \cdot \cos \pi = \frac{1}{2} m (v_f^2 - v_c^2) \Rightarrow d = \frac{v_c^2}{2\mu_k g} = 5.68 \text{ m.}$$

- 2) Velocity v_f

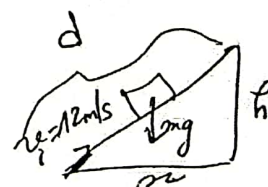
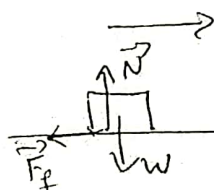
$$\text{like question 1} \Rightarrow v_f = \sqrt{v_c^2 - 2\mu_k g d} = 3.49 \text{ m/s.}$$

- 3) the height h

$$\text{we have } W = \vec{F} \cdot \vec{d} \text{ and } \vec{F} = -mg\vec{j}$$

$$\vec{d} = x\vec{i} + h\vec{j}$$

$$\Rightarrow W = -mg h = \frac{1}{2} m (v_f^2 - v_c^2) \Rightarrow h = \frac{v_c^2 - v_f^2}{2g} = 7.2 \text{ m.}$$



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Ex 6

Work Energy theorem:

the work is the change in kinetic energy.

$$W = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 = -\frac{1}{2} m v_i^2$$

and $W = (\sum \vec{F}) \cdot \vec{L} = (\vec{C}_{||} + \vec{C}_{\perp} + \vec{W}) \cdot \vec{L}$ / angle = 0
if $\cos = 1$

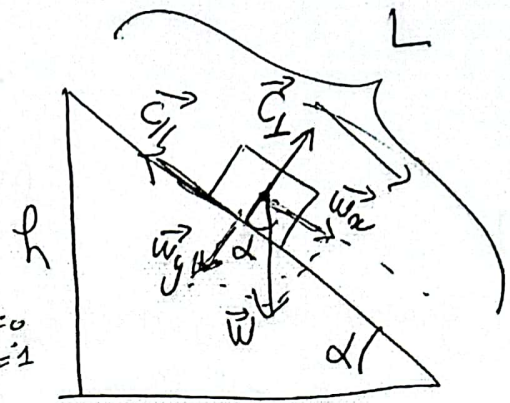
we have $\begin{cases} C_{\perp} = W_y = mg \cos \alpha \text{ and } \vec{C}_{\perp} = -\vec{W}_y \\ C_{||} = \mu_k \cdot C_{\perp} = \mu_k mg \cos \alpha \\ W_x = mg \sin \alpha \end{cases} \quad \sin \alpha = \frac{h}{L}$

$$\Rightarrow \underbrace{\frac{h}{\sin \alpha} (mg \sin \alpha - \mu_k mg \cos \alpha)}_{\text{work}} = \underbrace{-\frac{1}{2} m v_i^2}_{\text{work}}$$

$$\Rightarrow v_i^2 = (-2hg + \mu_k hg \frac{\cos \alpha}{\sin \alpha})$$

$$\Rightarrow v_i = \sqrt{2hg \left(\frac{\mu_k}{\tan \alpha} - 1 \right)}$$

Work



$\vec{C}_{||}$: friction force
 $\vec{C}_{\perp}$: Normal force
 $\vec{W} = \vec{W}_x + \vec{W}_y$: weight

Ex 7

a) moment of inertia of the system

$$I = \sum m_i r_i^2$$

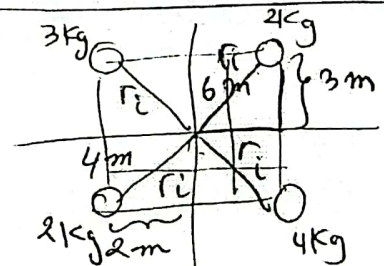
r_i is the distance of each particle from g-axis and all

particles have same distance $r_i = \sqrt{2^2 + 3^2} = 3.6 \text{ m}$.

$$I_{\text{sys}} = (2+3+4+2)(3.6)^2 = 142.56 \text{ kg} \cdot \text{m}^2$$

b) Rotational kinetic energy of system

$$E_{\text{rot}} = \frac{1}{2} I \omega^2 = \frac{1}{2} (142.56) (6)^2 = 2566.08 \text{ J}$$

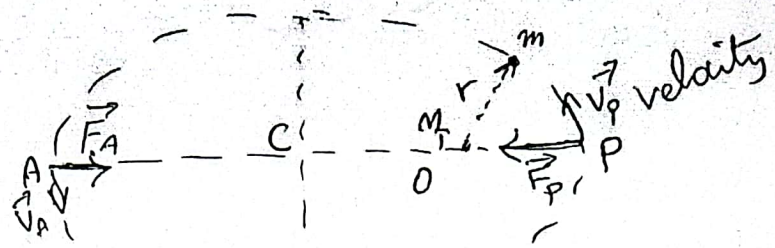


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E_{x08}

Work and Energy



1)

gravitational force $F_A < F_P$ because $OP < OA$.

$$F_A = G \frac{M_T m}{OA^2} < F_P = G \frac{M_T m}{OP^2}$$

2) i) $E_t = -G \frac{M_T m}{2a}$ $2a = \frac{AP}{2}$
(total) mechanical energy.

a) Satellite potential energy: $F = \frac{dE_p}{dr} \Rightarrow dE_p = F dr \Rightarrow E_p = \int F dr$

$$E_p = \int \frac{G M_T m}{r^2} dr = -\frac{G M_T m}{r} \quad \boxed{E_p = -\frac{G M_T m}{r}}$$

b) Kinetic energy E_k

$$E_t = E_k + E_p \Rightarrow E_k = E_t - E_p = -\frac{G M_T m}{2a} - \left(-\frac{G M_T m}{r}\right)$$

$$\boxed{E_k = G M_T m \left(\frac{1}{r} - \frac{1}{2a} \right)}$$

Velocity: $E_k = \frac{1}{2} m v^2 \Rightarrow v = \sqrt{\frac{2 E_k}{m}} = \sqrt{\frac{2}{m} \left(\frac{1}{2} G M_T m \left(\frac{2}{r} - \frac{1}{a} \right) \right)}$

$$\boxed{v = \sqrt{G M_T \left(\frac{2}{r} - \frac{1}{a} \right)}}$$

C - velocity at point A: $OA = r = 41400 \text{ km} = 41400 \times 10^3 \text{ m}$

$$OP = 6750 \text{ km} = 6750 \times 10^3 \text{ m}$$

$$2a = AP = OA + OP = (41400 + 6750) \times 10^3 = 48150 \times 10^3 \text{ m}$$

$$G M_T = 4,022 \times 10^{14} \text{ m}^3/\text{s}^2$$

$$v = G M_T \left(\frac{2}{r} - \frac{1}{a} \right) = 4,022 \times 10^{14} \left(\frac{2}{41400} - \frac{1}{48150} \right) \times 10^{-3}$$

$$v = \dots \dots \dots (\text{m/s})$$

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Work and Energy:

$$1 \text{ cm} \rightarrow 0,5 \text{ N}$$

velocity

1) velocity at point B:

the mobile has a potential energy at point A (at rest) and at any point on a circle the total energy is conserved, so:

$$E_{K(A)} + E_{P(A)} = E_{K(B)} + E_{P(B)}$$

$$\Rightarrow mgR = \frac{1}{2}mv_B^2 + mg(R - R\cos 45^\circ)$$

$$v_B^2 = 2mgR\cos 45^\circ \Rightarrow v_B = \sqrt{2gR\cos 45^\circ} = 3,74 \text{ m/s}$$

2) the exerted force are the weight and the contact force (Normal force)

$$\vec{w} = m \times g = 0,2 \times 10 = 2 \text{ N} \quad / \text{ Normal force } \vec{N} \text{ or } \vec{C}_1$$

$$N = C_1 = mg\cos 45^\circ = 1,4 \text{ N}$$

3) acceleration a_t and a_n / $a_n = \frac{v_B^2}{R} = \frac{(3,74)^2}{1} = 13,98 \text{ m/s}^2$

$$a_t: \Sigma F = ma \Rightarrow mg\sin 45^\circ = ma_t \Rightarrow a_t = g\sin 45^\circ = 10 \times 0,7 = 7 \text{ m/s}^2$$

4) The maximum compression of the spring:

$$\frac{1}{2}kx_{\max}^2 = \frac{1}{2}mv_D^2 + mgh_0 \quad (\text{at the bottom } h=0)$$

$$\Rightarrow x_{\max} = \sqrt{\frac{mv_D^2}{k}} = v_D \sqrt{\frac{m}{k}} \quad v_D = ?$$

$$E_{K(D)} + E_{P(D)} = E_{K(B)} + E_{P(B)} \Rightarrow E_{K(D)} = \frac{1}{2}mv_B^2 + mg(R - R\cos 45^\circ)$$

$$\frac{1}{2}mv_D^2 = \frac{1}{2}mv_B^2 + 2mg(R - R\cos 45^\circ)$$

$$v_D = \sqrt{2 \left(\frac{1}{2}v_B^2 + g(R - R\cos 45^\circ) \right)} = \sqrt{v_B^2 + 2g(R - R\cos 45^\circ)}$$

$$v_D = \sqrt{13,98 + 6} = 4,46 \text{ m/s}$$

$$x_{\max} = v_D \sqrt{\frac{m}{k}} = 4,46 \sqrt{\frac{0,2}{104}} = 0,02 \text{ m}$$

tan 45 = 2021 (NHSM)